HIERO PLATFORM ECOSYSTEM

FULL STACK ARCHITECTURE

Comprehensive Technical Stack Specification & Cross-Repository Analysis

Date: September 2026
Workspaces: 3 Unified Repositories
Status: Production Ready

1 Ecosystem Overview & Portal Breakdown

Final-Hiero
B2C Career & AI Suite

Student-facing core platform featuring AI Resume Builder, Real-time Mock Interviews, ATS Analyzer, and Multi-Service Gateway.

Bidge-Portal
B2B Academia Gateway

Institutional placement cell management, student batch record indexing, CSV upload ingestion, and role-based coordinator controls.

connect-portal-
B2B Industry & Recruiter

Recruiter matching engine, candidate pipeline management (Kanban DnD), job publishing, and applicant tracking interface.

2 Inter-System Network & Gateway Topology

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// HIERO Ecosystem Unified Topology & Port Distribution
┌──────────────────────────────────────────────────────────────────────────────────────────────┐ | [Final-Hiero] (Port :2816) ── Unified Express Gateway / Reverse Proxy & Static Host | | ├── AI Engine: Google Gemini (@google/generative-ai) + Groq Cloud LLaMA/Mixtral | | ├── Voice AI: Deepgram Speech-to-Text WebSocket + Tesseract.js OCR Pipeline | | ├── Rendering: Puppeteer Headless PDF Engine + PDFKit / Mammoth DOCX Exporter | | └── Security & DB: MongoDB Atlas (Mongoose) + Passport (Google/GitHub OAuth) + JWT | └──────────────────────────────────────────────────────────────────────────────────────────────┘ ▲ ▲
REST Sync / JWT Tokens Recruiter Data Link ▼ ▼ ┌──────────────────────────────────┐
┌──────────────────────────────────┐ | [Bidge-Portal] (Port :2410) | | [connect-portal-] (Port :2004) | | ├── Academic & Placement Gateway | | ├── Industry & Recruiter Dashboard | | ├── Express.js REST API Layer | | ├── React 18 + TypeScript + Vite | | ├── CSV-Parser Bulk Student Importer | | ├── Tailwind CSS + Lucide Icons | | └── Role-based Coordinator Controls | | └── Zustand State + React-DnD Kanban | └──────────────────────────────────┘ └──────────────────────────────────┘
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3 Cross-Repository Technology Comparison

ARCHITECTURE LAYER	FINAL-HIERO (B2C MAIN)	BIDGE-PORTAL (ACADEMIA)	CONNECT-PORTAL-(RECRUITER)
Primary Focus	AI Resume, Mock Interview, PWA & Gateway	College Placements, Student Ingestion	Recruiter ATS, Hiring Pipelines, Matching
Frontend Core	Vanilla HTML5, JS (ES6+), CSS3 (PWA)	Semantic HTML5, Vanilla JS Controllers	React 18.2.0 + TypeScript 5.3.3 (Vite 5)
UI & Styling	Custom CSS3, Glassmorphism, Keyframes	Custom Design System (style.css), Particles	Tailwind CSS 3.4.1 + PostCSS + Lucide Icons
State & Routing	Native JS State + Gateway URL Routing	Modular State (state.js) + REST Routing	Zustand 4.5.0 + React Router DOM v6
Backend Framework	Node.js + Express 4.22 (Unified Gateway)	Node.js + Express 4.19 (REST Gateway)	Node.js Static Server (server.cjs / serve)
AI / LLM Engines	Gemini AI, Groq SDK, Deepgram, Tesseract	Delegates via REST to Final-Hiero	Internal Matching Heuristics / REST API
Document Processing	Puppeteer (PDF), PDFKit, pdf-parse, Mammoth	csv-parser (Batch Import), Multer uploads	Client-side JSON / File DTOs
Database & Auth	MongoDB (Mongoose) + Passport OAuth + JWT	File Storage (JSON/CSV) + Express Auth	Zustand Persist + LocalStorage Tokens
Mobile Support	Capacitor (Android / Hybrid App packaging)	Responsive Viewport & Touch Layouts	Fully Responsive Recruiter Web UI

HIERO PLATFORM ECOSYSTEM

DEEP-DIVE MODULE BREAKDOWN

Detailed Package Inventories, Frameworks, and Specialized Libraries

4 Deep Dive: Final-Hiero Full Stack Inventory

Final-Hiero Core Capabilities & Packages

`Gateway Port 2816`

- ✓ **AI & Vision:** Google Gemini API (`@google/generative-ai`), Groq SDK (`groq-sdk`), Tesseract OCR (`tesseract.js`), Deepgram Audio.
- ✓ **Authentication & Security:** Passport.js (`passport-google-oauth20`, `passport-github2`), `jsonwebtoken` (JWT), `bcryptjs`, DNS SRV resolver.
- ✓ **Mobile & PWA:** Native Android wrapper using `capacitor.config.json`, Service Worker caching (`sw.js`), Web App Manifest.
- ✓ **Document Engine:** Puppeteer Chrome headless rendering, `pdfkit`, `pdf-parse`, `mammoth` (Word Docx), `html-to-docx`.
- ✓ **Monetization & Outreach:** Razorpay payment gateway integration (`razorpay`), Nodemailer email dispatch (`nodemailer`).
- ✓ **Database & Persistence:** MongoDB Atlas via `mongoose` (v7/v8) with resilient failover to JSON user tracking stores.

KEY DEPENDENCIES:

`@google/generative-ai: ^0.24.1` `groq-sdk: ^1.1.1` `tesseract.js: ^5.1.1` `puppeteer: ^24.37.5` `pdfkit: ^0.13.0` `pdf-parse: ^1.1.1` `mammoth: ^1.11.0` `express: ^4.22.1` `mongoose: ^7.6.3` `passport: ^0.6.0` `razorpay: ^2.9.2` `nodemailer: ^6.9.7` `winston: ^3.11.0` `multer: ^1.4.5` `uuid: ^9.0.0`

Bidge-Portal

`Port 2410`

High-throughput Academic Placement Portal connecting institutions with placement drives.

- ✓ **Server:** Express.js Unified Gateway hosting REST & UI.
- ✓ **CSV Engine:** `csv-parser` for multi-thousand student record ingestion.
- ✓ **Multi-Role JS:** Modular controllers for Admin, Coordinator, Student, and Sync.
- ✓ **Visuals:** Custom canvas interactive particle banner (`particles.js`).

`express: ^4.19.2` `csv-parser: ^3.0.0` `multer: ^1.4.5` `cors: ^2.8.5` `dotenv: ^16.4.5`

connect-portal-

`Port 2004 / SPA`

Modern React 18 & TypeScript recruiter suite designed for matching candidates with live openings.

- ✓ **Framework:** React 18.2 with TypeScript 5.3 & Vite 5.1 build tool.
- ✓ **Kanban Pipelines:** React DnD (`react-dnd`) for drag-and-drop hiring stages.
- ✓ **State & Auth:** Zustand store with persistent auth & application filters.
- ✓ **Design System:** Tailwind CSS 3.4 + Lucide React icon suite.

`react: ^18.2.0` `typescript: ^5.3.3` `vite: ^5.1.0` `zustand: ^4.5.0` `react-dnd: ^16.0.1` `tailwindcss: ^3.4.1` `lucide-react: ^0.344.0`

5 Architectural Conclusions & Strengths

High Performance & Low Latency

Groq SDK and Google Gemini provide sub-second AI inference, while Vite + Tailwind enable instantaneous UI rendering.

Modular Separation of Concerns

B2C students, B2B colleges, and B2B recruiters run on isolated gateways, avoiding monolithic bottlenecks.

Multi-Channel Deployment

Seamless transition between Web (PWA), Android (Capacitor), and Desktop across Linux / Windows servers.